
PE-EC802B: Industrial Automation and Control

Quick-Recall One-Liners Revision Guide

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Chapter 1: Sensors & Actuators

1. **Active Transducer definition** → Generates electrical output directly without external power source.
2. **Passive Transducer definition** → Requires external electrical source to measure parameter variations.
3. **Active transducer examples** → Thermocouple, piezoelectric sensor, solar cell.
4. **Passive transducer examples** → LVDT, RTD, thermistor, strain gauge.
5. **LVDT full form** → Linear Variable Differential Transformer.
6. **LVDT working principle** → Mutual inductance.
7. **LVDT null voltage cause** → Stray capacitances, harmonics, and magnetic leakage.
8. **LVDT phase shift across null position** → 180° phase shift.
9. **Thermocouple working principle** → Seebeck effect.
10. **Seebeck effect definition** → Temperature differences at joined dissimilar metals generate voltage.
11. **Type K thermocouple materials** → Chromel (positive) and Alumel (negative).
12. **Type J thermocouple materials** → Iron (positive) and Constantan (negative).
13. **Thermocouple Cold Junction Compensation necessity** → Corrects for fluctuations in reference junction temperature.
14. **Ultrasonic sensor distance formula** → $d = \frac{v \times t}{2}$.
15. **Piezoelectric actuator working principle** → Piezoelectric effect.
16. **Pneumatic actuator working fluid** → Compressed air.
17. **Electronic to pneumatic converter (I/P) input/output standards** → Converts 4–20 mA current to 3–15 psi pressure.
18. **Pneumatic to current converter (P/I) input/output standards** → Converts 3–15 psi pressure to 4–20 mA current.
19. **Components needed for a 4–20 mA loop** → Power supply, transmitter, and wire.
20. **Standard SI unit of torque** → Newton meter ($\text{N} \cdot \text{m}$).
21. **Displacement sensor measuring via plate separation or dielectric** → Capacitance sensor.
22. **Potentiometer primary measurement target** → Linear displacement.
23. **Control valve sizing design goals** → Avoids undersizing, oversizing, and excess cost.
24. **Smart sensor output style** → Digital.
25. **Radio Frequency Interference acronym** → RFI.

26. **Actuator type preferred for slow 10s time constant processes** → Piston and cylinder.

Chapter 2: Controller Tuning

1. **Proportional (P) controller equation** → $p(t) = K_c e(t) + p_s$.
2. **P-controller main disadvantage** → Steady-state error (offset) persists after load changes.
3. **Controller with maximum steady-state offset** → P-controller.
4. **Integral (I) controller equation** → $p(t) = \frac{1}{\tau_I} \int_0^t e(t^*) dt^* + p(0)$.
5. **I-controller main advantage** → Completely eliminates steady-state offset.
6. **I-controller main disadvantages** → Introduces phase lag and susceptible to integral windup.
7. **Derivative (D) controller equation** → $p(t) = \tau_D \frac{de(t)}{dt}$.
8. **Reason D-control cannot be used alone** → Output is zero under constant error.
9. **Relationship between Proportional Band (PB) and Gain (K_c)** → $K_c = \frac{100}{PB}$.
10. **Proportional Band expression units** → Percentage.
11. **Offset definition in control theory** → Steady-state error between setpoint and process variable.
12. **Op-amp PI controller feedback path components** → Resistor in series with capacitor.
13. **Op-amp PD controller feedback path components** → Resistor in parallel with capacitor.
14. **Parallel electronic PID design minimum op-amp count** → 2 operational amplifiers.
15. **Ziegler-Nichols (Z-N) tuning definition** → Experimental method determining gain factors via critical oscillation.
16. **Ziegler-Nichols parameter determination technique** → Graphical solution method.
17. **In direct-acting controller (CA4 definition), output increases when process measurement** → Decreases.
18. **In reverse-acting controller, output increases when process measurement** → Decreases.
19. **Action to reduce overshoots in a PID controller** → Increase the derivative time constant.
20. **PID controller selection criteria** → Offset elimination, small system changes, fast recovery.

Chapter 3: Automation Systems

1. **PLC full form** → Programmable Logic Controller.
2. **PLC advantages over electromagnetic relays** → High flexibility, online diagnostics, small space footprint.
3. **Four core PLC hardware modules** → CPU, Memory, Power Supply, I/O modules.
4. **Steps in a PLC scan cycle** → Read inputs, execute program, diagnostics/communication, write outputs.
5. **PLC scan time definition** → Time taken to execute the entire program.
6. **IEC 61131-3 graphical relay-like language** → Ladder Diagram (LD).
7. **IEC 61131-3 text-based high-level language** → Structured Text (ST).
8. **IEC 61131-3 assembly-like language** → Instruction List (IL).
9. **PLC noise immunity to electrical noises** → Excellent.

10. **SCADA full form** → Supervisory Control and Data Acquisition.
11. **Four core functions of SCADA** → Data acquisition, supervisory control, HMI display, alarm archiving.
12. **MTU full form in SCADA** → Master Terminal Unit.
13. **RTU full form in SCADA** → Remote Terminal Unit.
14. **HMI full form in SCADA** → Human Machine Interface.
15. **DCS full form** → Distributed Control Systems.
16. **PLC vs. DCS control scope** → PLC handles local discrete logic; DCS manages plant-wide continuous processes.
17. **PLC Instruction turning output ON/OFF after preset time** → Timer On Delay (TON).
18. **Instruction maintaining timer count on loss of rung power** → Retentive Timer (RTO).
19. **Non-PLC manufacturer in typical list** → Microsoft.
20. **Open-loop control system simple everyday example** → Automatic toaster.

Chapter 4: Advanced Control Techniques

1. **Feedforward control action style** → Proactive/anticipatory control based on measured disturbances.
2. **Feedforward control main advantage** → Rejects disturbances before process variable deviates.
3. **Feedforward control main disadvantage** → Only handles measured disturbances, requires precise model.
4. **Cascade control loop configuration** → Two nested feedback loops (primary outer, secondary inner).
5. **Cascade primary controller role** → Measures primary variable and outputs secondary controller setpoint.
6. **Cascade secondary controller role** → Measures intermediate variable and drives control valve.
7. **Cascade control primary benefit** → Rejects secondary-path disturbances before they affect primary output.
8. **Ratio control goal** → Maintain flow rate of one stream in fixed proportion to another.
9. **Wild flow definition in ratio control** → Uncontrolled stream whose flow rate fluctuates freely.
10. **Controlled flow calculation formula** → $F_{controlled} = R \times F_{wild}$.
11. **Override (selective) control purpose** → Protective control protecting equipment using High/Low selectors.
12. **Split-Range control definition** → Single controller output split to operate multiple valves.
13. **Split-range jacketed reactor example ranges** → 0–50% opens cooling; 50–100% opens heating.
14. **Adaptive control definition** → System adjusting controller parameters in real-time for changing dynamics.
15. **STR full form in adaptive control** → Self-Tuning Regulator.
16. **MRAC full form in adaptive control** → Model Reference Adaptive Control.

17. **STR vs. MRAC model parameters estimation** → STR estimates process parameters online; MRAC adjusts gains directly.
18. **IMC full form** → Internal Model Control.
19. **IMC core operating concept** → Embeds process model in parallel with physical process.
20. **Ideal IMC controller design** → Mathematical inverse of process model.